Indices, Surds & Quadratics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

The points of intersection satisfy both equations. Therefore, we can set the equations equal to each other: $2x + c = -x^2 + 4x + 5$ Rearranging this equation gives: $x^2 - 2x + (c - 5) = 0$ We are given that the graphs intersect at $x = -1$. Substituting $x = -1$ into the quadratic equation $y = -x^2 + 4x + 5$: $y = -(-1)^2 + 4(-1) + 5 = -1 - 4 + 5 = 0$ So, the point of intersection is $(-1, 0)$. Since this point also lies on the line $y = 2x + c$, we can substitute $x = -1$ and $y = 0$ into the linear equation.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The points A, B, and C have coordinates A(2, 5), B(k, 1), and C(8, -3). If A, B, and C are collinear, find the value of k.

For three points A, B, and C to be collinear, the gradient of the line segment AB must be equal to the gradient of the line segment BC (or AC). The gradient formula between two points (x_1, y_1) and (x_2, y_2) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$. Gradient of AB, $m_{AB} = \frac{1-5}{k-2} = \frac{-4}{k-2}$. Gradient of BC, $m_{BC} = \frac{-3-1}{8-k} = \frac{-4}{8-k}$. Equating the gradients: $\frac{-4}{k-2} = \frac{-4}{8-k}$ Since the numerators are equal and non-zero, the denominators.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

The original points are A(1, 2), B(4, 2), and C(2, 5). The curve $y = f(x)$ is a parabola. Since A and B have the same y-coordinate, the axis of symmetry is $x = \frac{1+4}{2} = 2.5$. The equation of the parabola can be written as $y = a(x - 2.5)^2 + k$. Substituting C(2, 5): $5 = a(2 - 2.5)^2 + k \Rightarrow 5 = 0.25a + k$. Step 1: Reflection in the line $y = x$. This transformation swaps the x and y coordinates. A'(2, 1), B'(2, 4), C'(5, 2). Step 2: Translation by the vector $(-3; 0)$. This subtracts 3.

© ClipSAT — clipsat.org | All Rights Reserved

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graph of the function $y = f(x)$. The coordinates of the points A, B, and C are given. The graph is transformed by a translation of 3 units to the right and 2 units down. What are the new coordinates of point A?

The original coordinates of point A are (-4, 2). A translation of 3 units to the right means we add 3 to the x-coordinate. A translation of 2 units down means we subtract 2 from the y-coordinate. New x-coordinate = Original x-coordinate + 3 = -4 + 3 = -1 New y-coordinate = Original y-coordinate - 2 = 2 - 2 = 0 Therefore, the new coordinates of point A are (-1, 0).

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the quadratic equation $x^2 - 5x + 6 = 0$, find the sum of the roots.

For a quadratic equation in the form $ax^2 + bx + c = 0$, the sum of the roots is given by the formula $-\frac{b}{a}$. In the given equation $x^2 - 5x + 6 = 0$, we have $a = 1$, $b = -5$, and $c = 6$. Therefore, the sum of the roots is $-\frac{-5}{1} = 5$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.

The point of intersection of two lines is the solution to the system of simultaneous equations that represent these lines. By visually inspecting the graph, we can identify the coordinates where the two lines cross. The first line appears to pass through the points (0, 3) and (3, 0). The second line appears to pass through the points (0, 1) and (2, 3). Let's find the equations of the lines. For the first line: Gradient $m_1 = \frac{0-3}{3-0} = \frac{-3}{3} = -1$. Using the point-slope form with (3, 0).

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The weights of bags of sugar are normally distributed with a mean of 1000 g and a standard deviation of 5 g. Let X be the weight of a randomly selected bag of sugar. Calculate the probability that a bag of sugar weighs between 990 g and 1010 g.

Let X be the weight of a randomly selected bag of sugar. We are given that $X \sim N(\mu = 1000, \sigma^2 = 5^2)$. We want to find $P(990 < X < 1010)$. First, we convert the weights to z-scores: For $X = 990$, $z = \frac{990-1000}{5} = \frac{-10}{5} = -2$. For $X = 1010$, $z = \frac{1010-1000}{5} = \frac{10}{5} = 2$. So we need to find $P(-2 < Z < 2)$, where Z is the standard normal distribution. This probability can be found using a standard normal distribution table or calculator. The area to the left of $z = 2$ is.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a sector of a circle with centre O. The arc length of the sector is given by 6π cm and the area of the sector is 27π cm^2. The radius of the circle is r cm and the angle of the sector is θ radians. Find the value of r and θ.

The formula for the arc length of a sector is $L = r\theta$, where L is the arc length, r is the radius, and θ is the angle in radians. We are given that $L = 6\pi$. So, we have the equation: $r\theta = 6\pi$ (1) The formula for the area of a sector is $A = \frac{1}{2}r^2\theta$, where A is the area, r is the radius, and θ is the angle in radians. We are given that $A = 27\pi$. So, we have the equation: $\frac{1}{2}r^2\theta = 27\pi$ (2) We can rewrite equation (2) as $r(r\theta) = 54\pi$. Sub.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point P.

To find the gradient of the tangent to the curve at $P(3, 2)$, we determine the derivative $\frac{dy}{dx}$ and evaluate it at $x = 3$. 1. Differentiate $y = x^2 - 4x + 5$ with respect to x using the power rule: $\frac{dy}{dx} = 2x - 4$ 2. Substitute $x = 3$ into the derivative: $\frac{dy}{dx}\Big|_{x=3} = 2(3) - 4 = 6 - 4 = 2$ Common errors: - 6 occurs if only the first term is differentiated (giving $2x$) and evaluated at $x = 3$, omitting the deriv.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A discrete random variable, X, can take the values 1, 2, 3, 4, and 5. The probability distribution of X is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4, 5\}$, where k is a constant. Find the value of k.

The sum of probabilities for all possible values of a discrete random variable must equal 1. Therefore, $\sum_{x=1}^{5} P(X = x) = 1$. Given $P(X = x) = kx$, we have: $P(X = 1) = k(1) = k$ $P(X = 2) = k(2) = 2k$ $P(X = 3) = k(3) = 3k$ $P(X = 4) = k(4) = 4k$ $P(X = 5) = k(5) = 5k$ Summing these probabilities: $k + 2k + 3k + 4k + 5k = 1$ $15k = 1$ $k = \frac{1}{15}$ Therefore, the value of k is $\frac{1}{15}$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD 6*cmandtheradiusis*USD3 cm, find the angle θ in radians.

The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is USD 2 radians.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A curve is given by the equation $y = x^3 - 3x^2 + 2x$. Find the gradient of the curve at $x = 2$.

To find the gradient of the curve, we need to find the first derivative of the equation with respect to x. The derivative of $y = x^3 - 3x^2 + 2x$ is $\frac{dy}{dx} = 3x^2 - 6x + 2$. To find the gradient at $x = 2$, substitute $x = 2$ into the derivative: $\frac{dy}{dx}|_{x=2} = 3(2)^2 - 6(2) + 2 = 3(4) - 12 + 2 = 12 - 12 + 2 = 2$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.

The coordinates of A are (2, 1) and the coordinates of C are (5, 5). Using the distance formula, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Length of AC = $\sqrt{(5-2)^2 + (5-1)^2}$ Length of AC = $\sqrt{(3)^2 + (4)^2}$ Length of AC = $\sqrt{9 + 16}$ Length of AC = $\sqrt{25}$ Length of AC = 5.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A fair coin is tossed 3 times. Let X be the number of heads obtained. What is the probability that X = 2?

This is a binomial distribution scenario. The number of trials is $n = 3$ (the number of coin tosses). The probability of success (getting a head) in a single trial is $p = \frac{1}{2}$, and the probability of failure (getting a tail) is $q = 1 - p = \frac{1}{2}$. We want to find the probability of getting exactly $k = 2$ heads. The binomial probability formula is $P(X = k) = \binom{n}{k} p^k q^{n-k}$. Substituting the values: $P(X = 2) = \binom{3}{2} (\frac{1}{2})^2 (\frac{1}{2})^{3-2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Differentiation & Integration

A particle's velocity v m/s at time t seconds is given by $v = \frac{1}{2}t^2 - 3t + 5$. The displacement of the particle from a fixed origin is s metres. It is known that at $t = 2$, the displacement $s = 7$. The displacement function $s(t)$ can be expressed in the form $s(t) = At^3 + Bt^2 + Ct + D$. Find the value of A and the value of D.

We are given the velocity function $v(t) = \frac{1}{2}t^2 - 3t + 5$. Since displacement s is the integral of velocity with respect to time, we have: $s(t) = \int v(t)\,dt = \int (\frac{1}{2}t^2 - 3t + 5)\,dt$ Integrating term by term, we get: $s(t) = \frac{1}{2} \cdot \frac{t^3}{3} - 3 \cdot \frac{t^2}{2} + 5t + K$ $s(t) = \frac{1}{6}t^3 - \frac{3}{2}t^2 + 5t + K$ Comparing this with the given form $s(t) = At^3 + Bt^2 + Ct + D$, we can identify $A = \frac{1}{6}$, $B = -\frac{3}{2}$, $C = 5$. The con.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

The first three terms of the binomial expansion of $(1 + kx)^n$ are $1 + 6x + 12x^2$. Find the value of k.

The binomial expansion of $(1 + x)^n$ is given by $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$. For the given expansion $(1 + kx)^n$, we can substitute $y = kx$ to get $(1 + y)^n$. The first three terms of the expansion are $1 + ny + \frac{n(n-1)}{2!}y^2 + \ldots$. Substituting $y = kx$ back, we have: $1 + n(kx) + \frac{n(n-1)}{2!}(kx)^2 + \ldots$ $1 + nkx + \frac{n(n-1)}{2!}k^2x^2 + \ldots$ We are given that the first three terms are $1 + 6x + 12x^2$. Comparing the coefficients: 1. Coefficient of x: $nk = 6$ 2. Coefficient of.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Functions: Composite & Inverse

A particle of mass 8 kg is moving under the action of a net force $\mathbf{F}$ N. The acceleration of the particle, $\mathbf{a}$ m/s^2, is given by the vector $\mathbf{a} = (2t - 4; t^2 - 9)$ for $t \geq 0$, where t is the time in seconds. The particle is initially at rest at the origin.

We are given the acceleration vector $\mathbf{a} = (2t - 4; t^2 - 9)$ and the mass of the particle $m = 8$ kg. By Newton's Second Law, the net force $\mathbf{F} = m\mathbf{a}$. Therefore, $\mathbf{F} = 8(2t - 4; t^2 - 9) = (16t - 32; 8t^2 - 72)$. The question asks for the magnitude of the net force at $t = 2$ seconds. Substituting $t = 2$ into the force vector.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Circular Measure

A sector of a circle with radius r cm has a perimeter of P cm. The area of the sector is A cm^2. Prove that $P^2 \geq 16A$.

Let the angle of the sector be θ radians. The arc length of the sector is $r\theta$. The perimeter $P = 2r + r\theta$. The area $A = \frac{1}{2}r^2\theta$. We need to prove $P^2 \geq 16A$. Substituting the expressions for P and A: $(2r + r\theta)^2 \geq 16\left(\frac{1}{2}r^2\theta\right)$ $r^2(2 + \theta)^2 \geq 8r^2\theta$ Since r^2 is positive, we can divide both sides by r^2: $(2 + \theta)^2 \geq 8\theta$ $4 + 4\theta + \theta^2 \geq 8\theta$ $\theta^2 - 4\theta + 4 \geq 0$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Series & Binomial Theorem

A particle of mass 5 kg moves under the action of a variable force $\mathbf{F}$ N. The force is given by $\mathbf{F} = (3x^2 - 2x)\mathbf{i}$ N, where x is the displacement in metres from a fixed origin O. The particle moves from $x = 2$ m to $x = 4$ m. Find the work done by the force $\mathbf{F}$ during this displacement.

The work done by a variable force $\mathbf{F}$ when a particle moves from position x_1 to x_2 is given by the integral $W = \int_{x_1}^{x_2} \mathbf{F} \cdot d\mathbf{r}$. In this case, the force is $\mathbf{F} = (3x^2 - 2x)\mathbf{i}$ and the displacement is along the x-axis, so $d\mathbf{r} = dx\mathbf{i}$. The particle moves from $x_1 = 2$ m to $x_2 = 4$ m. Therefore, the work done is: $W = \int_2^4 (3x^2 - 2x)dx$ Integrating term by term: $W = \left[\frac{3x^3}{3} - \frac{2x^2}{2}\right]_2^4$.

© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Kinematics

Solve the quadratic equation $2x^2 - 5x - 3 = 0$ by completing the square. State the vertex of the corresponding parabola $y = 2x^2 - 5x - 3$.

Factor 2 from the first two terms: $2\left(x^2 - \frac{5}{2}x\right) - 3 = 0$. Halve the coefficient of x: $\frac{1}{2}\left(-\frac{5}{2}\right) = -\frac{5}{4}$. Complete the square: $2\left(x - \frac{5}{4}\right)^2 - 2\left(\frac{25}{16}\right) - 3 = 0 \Rightarrow 2\left(x - \frac{5}{4}\right)^2 - \frac{25}{8} - \frac{24}{8} = 0 \Rightarrow 2\left(x - \frac{5}{4}\right)^2 = \frac{49}{8}$. Divide by 2: $\left(x - \frac{5}{4}\right)^2 = \frac{49}{16}$. Take square roots.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify $\left(\frac{8x^6}{27y^3}\right)^{1/3}$ completely.

a. $\frac{2x^2}{3y}$ ✓

b. $\frac{4x^2}{9y}$

c. $\frac{2x^3}{3y}$

d. $\frac{8x^2}{27y}$

Apply the power $1/3$ to each term: $8^{1/3} = 2$, $(x^6)^{1/3} = x^2$, $27^{1/3} = 3$, $(y^3)^{1/3} = y$. Result is $\frac{2x^2}{3y}$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify $\sqrt{48} - \sqrt{12} + \sqrt{75}$ into the form $k\sqrt{3}$.

a. $7\sqrt{3}$ ✓

b. $5\sqrt{3}$

c. $9\sqrt{3}$

d. $6\sqrt{3}$

$\sqrt{48} = 4\sqrt{3}$, $\sqrt{12} = 2\sqrt{3}$, $\sqrt{75} = 5\sqrt{3}$. Then $4\sqrt{3} - 2\sqrt{3} + 5\sqrt{3} = 7\sqrt{3}$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Rationalize the denominator of $\frac{6}{3 - \sqrt{3}}$ and write in simplest form.

a. $3 + \sqrt{3}$ ✓

b. $2 + \sqrt{3}$

c. $3 - \sqrt{3}$

d. $6 + 2\sqrt{3}$

Multiply numerator and denominator by $(3 + \sqrt{3})$: $\frac{6(3 + \sqrt{3})}{9 - 3} = \frac{6(3 + \sqrt{3})}{6} = 3 + \sqrt{3}$.

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $2x^2 - 12x + 11$ in the form $a(x-h)^2 + k$.

a. $2(x-3)^2 - 7$ ✓

b. $2(x-3)^2 + 11$

c. $2(x-6)^2 - 25$

d. $2(x-3)^2 - 11$

$2(x^2-6x)+11 = 2[(x-3)^2-9]+11 = 2(x-3)^2-18+11 = 2(x-3)^2-7$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the minimum value of $f(x) = 3x^2 + 18x + 20$ and the value of x at which it occurs.

a. **Minimum is** -7 **at** $x = -3$ ✓

b. Minimum is 20 at $x = 0$

c. Minimum is -7 at $x = 3$

d. Minimum is 7 at $x = -3$

$3(x^2+6x)+20 = 3[(x+3)^2-9]+20 = 3(x+3)^2-27+20 = 3(x+3)^2-7$. Minimum is -7 when $x = -3$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The curve $y = -(x+4)^2 + 15$ has a stationary point. State its nature and coordinates.

a. **Maximum at** $(-4, 15)$ ✓

b. Minimum at $(-4, 15)$

c. Maximum at $(4, 15)$

d. Minimum at $(4, -15)$

Since the leading coefficient is negative, the parabola opens downward, giving a maximum at $(-4, 15)$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the set of values of k for which the equation $x^2 + kx + (k+3) = 0$ has two distinct real roots.

a. $k < -2$ **or** $k > 6$ ✓

b. $-2 < k < 6$

c. $k \leq -2$ or $k \geq 6$

d. $k > 6$

Distinct real roots require $\Delta > 0$: $k^2 - 4(1)(k+3) > 0 \Rightarrow k^2 - 4k - 12 > 0 \Rightarrow (k-6)(k+2) > 0$. Thus $k < -2$ or $k > 6$.

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The line $y = 2x + c$ is tangent to the curve $y = x^2 - 4x + 7$. Find the exact value of c.

a. -2 ✓

b. 2

c. 7

d. -6

Set $x^2 - 4x + 7 = 2x + c \Rightarrow x^2 - 6x + (7 - c) = 0$. For tangency, $\Delta = 0$:

$36 - 4(7 - c) = 0 \Rightarrow 36 - 28 + 4c = 0 \Rightarrow 4c = -8 \Rightarrow c = -2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine the condition on p such that the quadratic $px^2 + 4x + p = 0$ has no real roots, where $p > 0$.

a. $p > 2$ ✓

b. $0 < p < 2$

c. $p > 4$

d. $p < -2$

No real roots means $\Delta < 0$: $16 - 4p^2 < 0 \Rightarrow 4p^2 > 16 \Rightarrow p^2 > 4$. Since $p > 0$, $p > 2$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the disguised quadratic equation $x^4 - 5x^2 - 36 = 0$ for all real solutions.

a. $x = \pm 3$ ✓

b. $x = \pm 2$

c. $x = \pm 3,\ \pm 2i$

d. $x = 9$

Let $u = x^2$: $u^2 - 5u - 36 = (u - 9)(u + 4) = 0$. Since x is real, $u = x^2 = 9 \Rightarrow x = \pm 3$ ($u = -4$ has no real solutions).

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A rectangular enclosure is formed using 80 meters of fencing against a straight stone wall. What is the maximum possible area of the enclosure in square meters?

a. $800\ \text{m}^2$ ✓

b. $400\ \text{m}^2$

c. $1600\ \text{m}^2$

d. $600\ \text{m}^2$

Let width be x. Length along wall is $80 - 2x$. Area $A(x) = x(80 - 2x) = 80x - 2x^2 = -2(x - 20)^2 + 800$. The maximum area is $800\ \text{m}^2$ at $x = 20$ m.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the discriminant of $3x^2 - 5x - 10 = 0$ and state the number of real roots.

a. $\Delta = -145$: two distinct real roots

b. $\Delta = 145$: one repeated real root

c. $\Delta = 153$: no real roots

d. $\Delta = 145$**: two distinct real roots ✓**

$\Delta = b^2 - 4ac = (-5)^2 - 4(3)(-10) = 145$, so the equation has two distinct real roots.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

a. **6 cm ✓**

b. 5 cm

c. 7 cm

d. 8 cm

In a right-angled triangle, the altitude to the hypotenuse creates three similar triangles. Specifically, triangle ABD is similar to triangle CAD, and both are similar to the original triangle ABC. By the geometric mean theorem (or by using similarity ratios), the square of the altitude to the hypotenuse is equal to the product of the two segments it divides the hypotenuse into. Therefore, $AD^2 = BD \times DC$. Given $BD = 4$ cm and $DC = 9$ cm. $AD^2 = 4 \times 9$ $AD^2 = 36$ $AD = \sqrt{36}$ $AD = 6$ cm.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a right-angled triangle ABC, with the right angle at C. The altitude from C to the hypotenuse AB is CD. Given that $AD = 4$ and $DB = 9$, find the length of the altitude CD.

a. **3 ✓**

b. 4

c. 5

d. 6

In a right-angled triangle, the altitude to the hypotenuse creates three similar triangles: $\Delta ABC \sim \Delta ACD \sim \Delta CBD$. Applying the geometric mean theorem for the altitude, we have $CD^2 = AD \times DB$. Given $AD = 4$ and $DB = 9$, we substitute these values into the formula: $CD^2 = 4 \times 9 = 36$. Taking the square root of both sides, $CD = \sqrt{36} = 6$. Therefore, the length of the altitude CD is 6.

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Rationalise the denominator of $\frac{1}{\sqrt{23}}$.

a. $\frac{\sqrt{23}}{23}$

b. $\frac{1\sqrt{23}}{1}$

c. $\frac{1\sqrt{23}}{23}$ ✓

d. $1\sqrt{23}$

Multiply top and bottom by $\sqrt{23}$: $\frac{1}{\sqrt{23}} \times \frac{\sqrt{23}}{\sqrt{23}} = \frac{1\sqrt{23}}{23}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Write $a^{-3} \times a^{-1}$ in the form a^k, giving the value of k.

a. **k=-4** ✓

b. k=-3

c. k=4

d. k=-2

Using the laws of indices, the exponents combine to give a^{-4}.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify $\sqrt{96}$, giving your answer in the form $a\sqrt{b}$ with b as small as possible.

a. $4\sqrt{24}$

b. $5\sqrt{6}$

c. $4\sqrt{6}$ ✓

d. $4\sqrt{7}$

$96 = 4^2 \times 6$, so $\sqrt{96} = \sqrt{4^2 \times 6} = 4\sqrt{6}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Circular Measure

The histogram below shows the distribution of heights of a group of students. The total number of students in the group is 50. If the frequency density for the class interval $60 \leq h < 70$ is 1.2, find the value of k.

a. 10

b. **12** ✓

c. 15

d. 20

The area of each bar in a histogram represents the frequency of the corresponding class interval. The frequency density is given by $\frac{\text{frequency}}{\text{class width}}$. The class interval $60 \leq h < 70$ has a class width of $70 - 60 = 10$. The frequency density for this interval is given as 1.2. Therefore, the frequency for this interval is frequency density $\times$ class width $= 1.2 \times 10 = 12$. The histogram shows that the frequency for the interval $60 \leq h < 70$ is represented by k..

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Series & Binomial Theorem

In triangle ABC, angle ABC = θ degrees, and angle ACB = 2θ degrees. The lengths of sides AB and AC are 10 cm and 15 cm respectively. The length of BC is denoted by x cm. Given that the area of triangle ABC is 60 cm^2, find the value of x, to 3 significant figures.

a. 13.7

b. 11.4

c. **12.8** ✓

d. 14.9

Let the angles of triangle ABC be denoted as follows: $\angle ABC = \theta$, $\angle ACB = 2\theta$. By the sine rule, $\frac{AC}{\sin(\theta)} = \frac{AB}{\sin(2\theta)}$. Substituting the given values, $\frac{15}{\sin(\theta)} = \frac{10}{\sin(2\theta)}$. Using the double angle identity, $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$, we get $\frac{15}{\sin(\theta)} = \frac{10}{2\sin(\theta)\cos(\theta)}$. Since $\sin(\theta) eq 0$ for a non-degenerate triangle, we can cancel $\sin(\theta)$ from both sides.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Kinematics

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is 6 cm and the radius is 3 cm, find the angle θ in radians.

a. $\frac{1}{2}$ radians

b. 2 **radians** ✓

c. $\frac{2}{3}$ radians

d. $\frac{3}{2}$ radians

The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is 2 radians.

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 12.01 Forces & Equilibrium

A sector has radius 6 cm and angle 0.3 rad. Find the arc length.

a. **9/5** ✓

b. 63/10

c. 14/5

d. 27/5

Applying Circular Measure principles, $s = r\theta = 6 \times 0.3 = 9/5$ cm Therefore, the verified correct option is 9/5.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 13.01 Newton's Laws

A sector OAB has radius 8 cm and angle $\angle AOB = 1.2$ rad. Find the area of the minor segment cut off by chord AB, leaving your answer in exact form.

a. $32(1.2 - \sin(1.2))$ ✓

b. $64(1.2 - \sin(1.2))$

c. $33(1.2 - \sin(1.2))$

d. $32(1.2 + \sin(1.2))$

Segment $=$ Sector $-$ Triangle $= \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta = 32(1.2 - \sin(1.2))$ cm^2

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 14.01 Energy, Work & Power

Find the equation of the line through $(0, -7)$ and $(-4, -5)$, in the form $y = mx + c$.

a. y=-1/2x+7

b. **y=-1/2x-7** ✓

c. y=-1/2x-6

d. y=1/2x-7

Using $y - y_1 = m(x - x_1)$: $y + 7 = -1/2(x - 0)$, which rearranges to $y = -1/2x - 7$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 15.01 Representation of Data

A circle has equation $(x - 7)^2 + (y - 5)^2 = 25$. State its centre and radius.

a. centre $(5, 7)$, radius 5

b. **centre** $(7, 5)$**, radius** 5 ✓

c. centre $(-7, -5)$, radius 5

d. centre $(7, 5)$, radius 25

Comparing to $(x - a)^2 + (y - b)^2 = r^2$: centre $(7, 5)$, and $r = \sqrt{25} = 5$.

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 16.01 Permutations & Combinations

A sector has radius 10 cm and angle 1.7 rad. Find its area.

a. 170

b. 95

c. **85** ✓

d. 17

$A = \frac{1}{2}r^2\theta = \frac{1}{2}(10)^2(1.7) = 85$ cm^2

© ClipSAT — clipsat.org | All Rights Reserved